Comparison of Feature Selection Across Models

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I was curious if any of the models favored the same features, especially since all models returned very similar MAEs. Because not all models perform feature selection directly, I used a comparison approach to identify features that appeared across multiple methods. The models evaluated were linear regression, LASSO, ridge, elastic net, and PLS. I prompted generative AI to help generate code for this comparison.

No features were prioritized across all models. However, 18 features were consistently identified across LASSO, Ridge, and Elastic Net: f111, f218, f221, f313, f383, f384, f398, f413, f471, f526, f533, f536, f615, f663, f673, f674, f775, and f80. Only five features (f113, f145, f199, f300, f720) were selected by LASSO alone.

Linear regression and PLS did not show overlap with other models. Linear regression appears to be unstable for feature ranking in this dataset, likely due to high feature correlation affecting coefficient estimates. PLS, on the other hand, constructs components from combinations of variables, so its notion of feature importance is not directly comparable to models that select features based on individual performance. While none of the top 20 features contributing to the six selected PLS components overlapped with the other models, this likely reflects a difference in how importance is distributed across combinations of variables rather than individual predictors.

It may be worthwhile to apply predictive analytics to both the LASSO and PLS data sets independently. They could both reveal important insight.

If we had more time, it would be worth exploring other loss functions or ways of evaluating feature importance.

Loading libraries

```r
library(caret)    # data splitting + nearZeroVar
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```r
library(glmnet)   # LASSO, Ridge, Elastic Net
```

```
## Loading required package: Matrix
```

```
## Loaded glmnet 4.1-10
```

```r
library(pls)      # PLS model
```

```
##
## Attaching package: 'pls'

## The following object is masked from 'package:caret':
##
##     R2

## The following object is masked from 'package:stats':
##
##     loadings
```

```r
library(dplyr)    # Cleaner data manipulation
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)  # Nicer plots
```

Set seed

```r
set.seed(123)
```

Loading Gabriel's cleaned train and test datasets

```r
## Set your own working directory
train <- read.csv("train_clean_final.csv")
test <- read.csv("test_clean_final.csv")
```

Creating training and validation partitions

```r
index <- createDataPartition(train$loss, p = 0.7, list = FALSE)

training <- train[index, ]
validation <- train[-index, ]
```

Removing near-zero variance features

```r
# Identify near-zero variance features from the training partition only
nzv <- nearZeroVar(training[, names(training) != "loss"], names = TRUE)

# Remove near-zero variance features from both training and validation
training_comp <- training[, !names(training) %in% nzv]
validation_comp <- validation[, !names(validation) %in% nzv]

dim(training_comp)
```

```
## [1] 56000   711
```

```r
dim(validation_comp)
```

```
## [1] 24000   711
```

Defining train and validate partitions

```r
# Define predictor matrices and target vectors for model comparison
x_train <- as.matrix(training_comp[, names(training_comp) != "loss"])
y_train <- training_comp$loss

x_val <- as.matrix(validation_comp[, names(validation_comp) != "loss"])
y_val <- validation_comp$loss
```

Models for comparison

```r
# --- Fit models for comparison ---

# LASSO
cv_lasso <- cv.glmnet(x_train, y_train, alpha = 1, type.measure = "mae")

# Ridge
cv_ridge <- cv.glmnet(x_train, y_train, alpha = 0, type.measure = "mae")

# Elastic Net already defined later (you can leave it or move it here)

# Linear Regression
lm_model <- lm(loss ~ ., data = training_comp)

# PLS
pls_model <- plsr(
  loss ~ .,
  data = training_comp,
  ncomp = 10,
  scale = FALSE,
  validation = "CV"
)

# Find best_k for PLS
msep <- MSEP(pls_model, estimate = "CV")
cv_mse <- as.vector(msep$val[1, 1, ])
best_k <- which.min(cv_mse)
```

Compare LASSO and Elastic Net

```r
# LASSO
lasso_coef <- coef(cv_lasso, s = "lambda.min")
lasso_features <- rownames(lasso_coef)[lasso_coef[,1] != 0]
lasso_features <- lasso_features[lasso_features != "(Intercept)"]

# Elastic Net (example alpha = 0.5)
cv_enet <- cv.glmnet(x_train, y_train, alpha = 0.5, type.measure = "mae")
enet_coef <- coef(cv_enet, s = "lambda.min")
enet_features <- rownames(enet_coef)[enet_coef[,1] != 0]
enet_features <- enet_features[enet_features != "(Intercept)"]

# Overlap
intersect(lasso_features, enet_features)
```

```
##  [1] "f5"   "f6"   "f53"  "f55"  "f65"  "f67"  "f68"  "f70"  "f71"  "f80"
## [11] "f82"  "f111" "f112" "f122" "f130" "f135" "f143" "f149" "f160" "f179"
## [21] "f180" "f190" "f200" "f218" "f221" "f223" "f232" "f238" "f241" "f251"
## [31] "f281" "f286" "f292" "f293" "f294" "f305" "f313" "f314" "f323" "f361"
## [41] "f374" "f383" "f384" "f398" "f403" "f405" "f413" "f422" "f428" "f451"
## [51] "f471" "f479" "f499" "f518" "f522" "f523" "f526" "f533" "f536" "f546"
## [61] "f556" "f588" "f601" "f604" "f609" "f612" "f614" "f615" "f616" "f620"
## [71] "f631" "f650" "f651" "f652" "f660" "f663" "f665" "f666" "f673" "f674"
## [81] "f676" "f680" "f704" "f705" "f721" "f725" "f734" "f746" "f747" "f752"
## [91] "f755" "f757" "f758" "f759" "f765" "f775"
```

Ridge feature importance

```r
ridge_coef <- coef(cv_ridge, s = "lambda.min")
ridge_top <- names(sort(abs(ridge_coef[,1]), decreasing = TRUE))[1:20]
```

PLS importance via loadings

```r
loadings <- pls_model$loadings[, 1:best_k]
importance <- rowSums(abs(loadings))
pls_top <- names(sort(importance, decreasing = TRUE))[1:20]
```

Top linear regression features by coefficient magnitude

```r
coefs <- coef(lm_model)
lm_top <- names(sort(abs(coefs), decreasing = TRUE))[1:20]
lm_top <- lm_top[lm_top != "(Intercept)"]
```

Comparing important features - LASSO, Elastic Net, Ridge, and PLS

```r
common <- Reduce(intersect, list(lasso_features, enet_features, ridge_top, pls_top))
common
```

```
## character(0)
```

Comparing important features - LASSO, Elastic Net, Ridge, PLS, and Linear Regression

```r
common <- Reduce(intersect, list(lasso_features, enet_features, ridge_top, pls_top, lm_top))
common
```

```
## character(0)
```

Finding important features that show up in three or more models

```r
all_features <- c(
  lasso_features,
  enet_features,
  ridge_top,
  pls_top,
  lm_top
)



feature_counts <- sort(table(all_features), decreasing = TRUE)
feature_counts <- feature_counts[names(feature_counts) != "(Intercept)"]
feature_counts
```

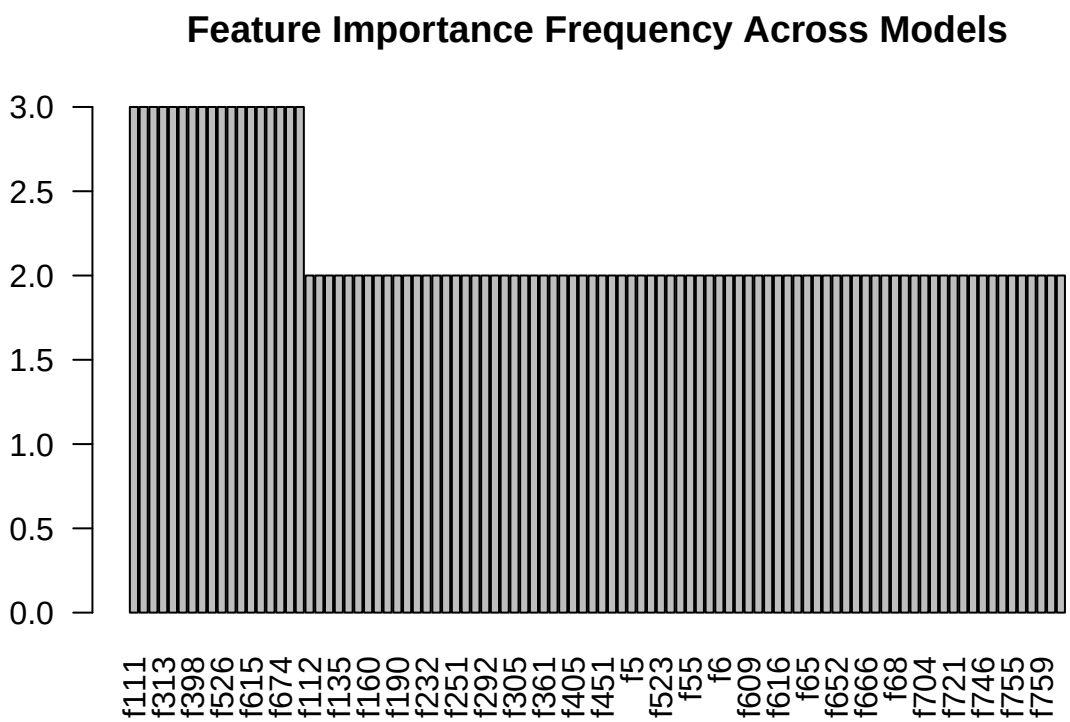
```
## all_features
## f111 f218 f221 f313 f383 f384 f398 f413 f471 f526 f533 f536 f615 f663 f673 f674
##    3    3    3    3    3    3    3    3    3    3    3    3    3    3    3    3
## f775  f80 f112 f122 f130 f135 f143 f149 f160 f179 f180 f190 f200 f223 f232 f238
##    3    3    2    2    2    2    2    2    2    2    2    2    2    2    2    2
## f241 f251 f281 f286 f292 f293 f294 f305 f314 f323 f361 f374 f403 f405 f422 f428
##    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2
## f451 f479 f499   f5 f518 f522 f523  f53 f546  f55 f556 f588   f6 f601 f604 f609
##    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2
## f612 f614 f616 f620 f631  f65 f650 f651 f652 f660 f665 f666  f67 f676  f68 f680
##    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2
##  f70 f704 f705  f71 f721 f725 f734 f746 f747 f752 f755 f757 f758 f759 f765  f82
##    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2
## f105 f106 f107 f108 f113 f115 f125 f126 f127 f128 f145 f199  f22  f23 f295 f296
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
## f300 f401 f459 f468 f475 f480 f481 f485 f493 f494 f495 f497 f501 f502 f507 f510
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
## f512 f530 f542 f547 f552 f557 f562 f564 f572 f581 f709 f720 f726 f769  f95  f96
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
##  f97  f98
##    1    1
```

```r
feature_counts[feature_counts >= 3]
```

```
## all_features
## f111 f218 f221 f313 f383 f384 f398 f413 f471 f526 f533 f536 f615 f663 f673 f674
##    3    3    3    3    3    3    3    3    3    3    3    3    3    3    3    3
## f775  f80
##    3    3
```

Visual model of shared features

```r
barplot(feature_counts[feature_counts >= 2],
        las = 2,
        main = "Feature Importance Frequency Across Models")
```



Finding important features that show up in four or more models

```r
all_features <- c(
  lasso_features,
  enet_features,
  ridge_top,
  pls_top,
  lm_top
)



feature_counts <- sort(table(all_features), decreasing = TRUE)
feature_counts <- feature_counts[names(feature_counts) != "(Intercept)"]
feature_counts
```

```
## all_features
## f111 f218 f221 f313 f383 f384 f398 f413 f471 f526 f533 f536 f615 f663 f673 f674
##    3    3    3    3    3    3    3    3    3    3    3    3    3    3    3    3
## f775  f80 f112 f122 f130 f135 f143 f149 f160 f179 f180 f190 f200 f223 f232 f238
##    3    3    2    2    2    2    2    2    2    2    2    2    2    2    2    2
```

```
## f241 f251 f281 f286 f292 f293 f294 f305 f314 f323 f361 f374 f403 f405 f422 f428
##    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2
## f451 f479 f499   f5 f518 f522 f523  f53 f546  f55 f556 f588   f6 f601 f604 f609
##    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2
## f612 f614 f616 f620 f631  f65 f650 f651 f652 f660 f665 f666  f67 f676  f68 f680
##    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2
##  f70 f704 f705  f71 f721 f725 f734 f746 f747 f752 f755 f757 f758 f759 f765  f82
##    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2    2
## f105 f106 f107 f108 f113 f115 f125 f126 f127 f128 f145 f199  f22  f23 f295 f296
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
## f300 f401 f459 f468 f475 f480 f481 f485 f493 f494 f495 f497 f501 f502 f507 f510
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
## f512 f530 f542 f547 f552 f557 f562 f564 f572 f581 f709 f720 f726 f769  f95  f96
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
##  f97  f98
##    1    1
```

```r
feature_counts[feature_counts >= 4]
```

```
## named integer(0)
```

Discovering which features selected by LASSO are also prioritized in two or more other models

```r
other_models <- c(enet_features, ridge_top, pls_top, lm_top)
lasso_overlap_counts <- table(other_models[other_models %in% lasso_features])
lasso_stable <- names(lasso_overlap_counts[lasso_overlap_counts >= 2])
lasso_stable
```

```
##  [1] "f111" "f218" "f221" "f313" "f383" "f384" "f398" "f413" "f471" "f526"
## [11] "f533" "f536" "f615" "f663" "f673" "f674" "f775" "f80"
```

Features selected by LASSO but only one other model

```r
# All LASSO features
lasso_all <- lasso_features

# Features that appear in >= 2 other models
lasso_stable <- names(lasso_overlap_counts[lasso_overlap_counts >= 2])

# LASSO features NOT supported by >=2 other models
lasso_unique_1 <- setdiff(lasso_all, lasso_stable)

lasso_unique_1
```

```
##  [1] "f5"   "f6"   "f53"  "f55"  "f65"  "f67"  "f68"  "f70"  "f71"  "f82"
## [11] "f112" "f113" "f122" "f130" "f135" "f143" "f145" "f149" "f160" "f179"
## [21] "f180" "f190" "f199" "f200" "f223" "f232" "f238" "f241" "f251" "f281"
## [31] "f286" "f292" "f293" "f294" "f300" "f305" "f314" "f323" "f361" "f374"
## [41] "f403" "f405" "f422" "f428" "f451" "f479" "f499" "f518" "f522" "f523"
## [51] "f546" "f556" "f588" "f601" "f604" "f609" "f612" "f614" "f616" "f620"
## [61] "f631" "f650" "f651" "f652" "f660" "f665" "f666" "f676" "f680" "f704"
## [71] "f705" "f720" "f721" "f725" "f734" "f746" "f747" "f752" "f755" "f757"
## [81] "f758" "f759" "f765"
```

Features selected by LASSO but not any other models

```r
# All LASSO features
lasso_all <- lasso_features

# Features that appear in >= 2 other models
lasso_stable <- names(lasso_overlap_counts[lasso_overlap_counts >= 1])

# LASSO features NOT supported by >=2 other models
lasso_unique_0 <- setdiff(lasso_all, lasso_stable)

lasso_unique_0
```

```
## [1] "f113" "f145" "f199" "f300" "f720"
```

Discovering which features used by PLS are also prioritized in two or more other models

```r
# Combine non-PLS model feature lists
other_model_features <- c(
  lasso_features,
  enet_features,
  ridge_top,
  lm_top
)

# Count how often each PLS top feature appears in the other models
pls_overlap_counts <- table(other_model_features[other_model_features %in% pls_top])

pls_overlap_counts
```

```
## < table of extent 0 >
```

```r
head(pls_top)
```

```
## [1] "f501" "f481" "f475" "f495" "f125" "f126"
```

```r
head(lasso_features)
```

```
## [1] "f5"  "f6"  "f53" "f55" "f65" "f67"
```

Determining which features are selected by both LASSO and each of the other models

```r
intersect(lasso_features, enet_features)
```

```
##  [1] "f5"   "f6"   "f53"  "f55"  "f65"  "f67"  "f68"  "f70"  "f71"  "f80"
## [11] "f82"  "f111" "f112" "f122" "f130" "f135" "f143" "f149" "f160" "f179"
## [21] "f180" "f190" "f200" "f218" "f221" "f223" "f232" "f238" "f241" "f251"
## [31] "f281" "f286" "f292" "f293" "f294" "f305" "f313" "f314" "f323" "f361"
## [41] "f374" "f383" "f384" "f398" "f403" "f405" "f413" "f422" "f428" "f451"
## [51] "f471" "f479" "f499" "f518" "f522" "f523" "f526" "f533" "f536" "f546"
## [61] "f556" "f588" "f601" "f604" "f609" "f612" "f614" "f615" "f616" "f620"
## [71] "f631" "f650" "f651" "f652" "f660" "f663" "f665" "f666" "f673" "f674"
## [81] "f676" "f680" "f704" "f705" "f721" "f725" "f734" "f746" "f747" "f752"
## [91] "f755" "f757" "f758" "f759" "f765" "f775"
```

```r
intersect(lasso_features, ridge_top)
```

```
##  [1] "f80"  "f111" "f218" "f221" "f313" "f383" "f384" "f398" "f413" "f471"
## [11] "f526" "f533" "f536" "f615" "f663" "f673" "f674" "f775"
```

```r
intersect(lasso_features, pls_top)
```

```
## character(0)
```

```r
intersect(lasso_features, lm_top)
```

```
## character(0)
```